

NexTwin AI

Edge-Powered Industrial Digital Twin Platform for Smart Manufacturing

Team NexTwin

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Tata Technologies InnoVent 2026

Problem Statement

Industries continue to rely on reactive monitoring systems, resulting in costly downtime, inefficient maintenance practices, and limited operational visibility.

- Unexpected Machine Failures
- Unplanned Downtime
- Energy Inefficiency
- Production Bottlenecks
- Lack of Real-Time Operational Visibility
- Delayed Maintenance Decisions

Impact of the Problem Statement

Operational inefficiencies directly affect productivity, profitability, and sustainability.

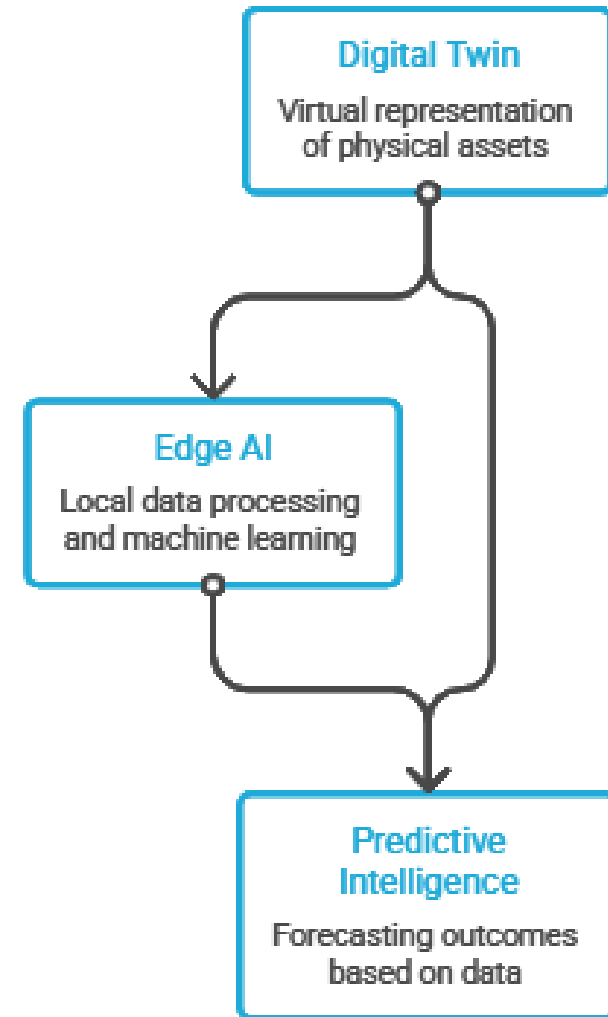
- Millions lost annually due to unplanned downtime and machine failures.
- 20–30% of maintenance activities are unnecessary, increasing operational costs.
- Energy inefficiencies lead to higher power consumption and reduced profitability.
- Lack of predictive insights results in delayed decision-making and lower productivity.
- Manufacturers need intelligent systems to improve visibility, efficiency, and reliability

Existing Solutions Are Fragmented

Existing System	Primary Purpose	Key Limitation
SCADA Systems	Real-time monitoring and control	Monitoring only; lacks predictive intelligence and failure forecasting.
ERP Platforms	Resource and business process management	Focuses on business operations but lacks machine-level analytics and real-time insights.
Traditional Dashboards	Visualization of operational metrics	Displays historical data but cannot predict failures, simulate scenarios, or recommend actions.

Introducing NexTwin AI

NexTwin AI transforms industrial operations by combining Edge AI, Digital Twins, Predictive Analytics, and Intelligent Decision Support into a unified platform. The system continuously monitors machine health, predicts failures, detects anomalies, optimizes energy consumption, simulates factory operations, and assists operators through an AI Copilot, enabling smarter, safer, and more efficient manufacturing.



Core Platform Overview

- Digital Twin
- Predictive Maintenance
- Anomaly Detection
- Energy Optimization
- Simulation Lab
- AI Copilot
- Reports Engine
- Smart Alerts



Digital Twin



Predictive Maintenance



Anomaly Detection



Energy Optimization



Simulation Lab



AI Copilot

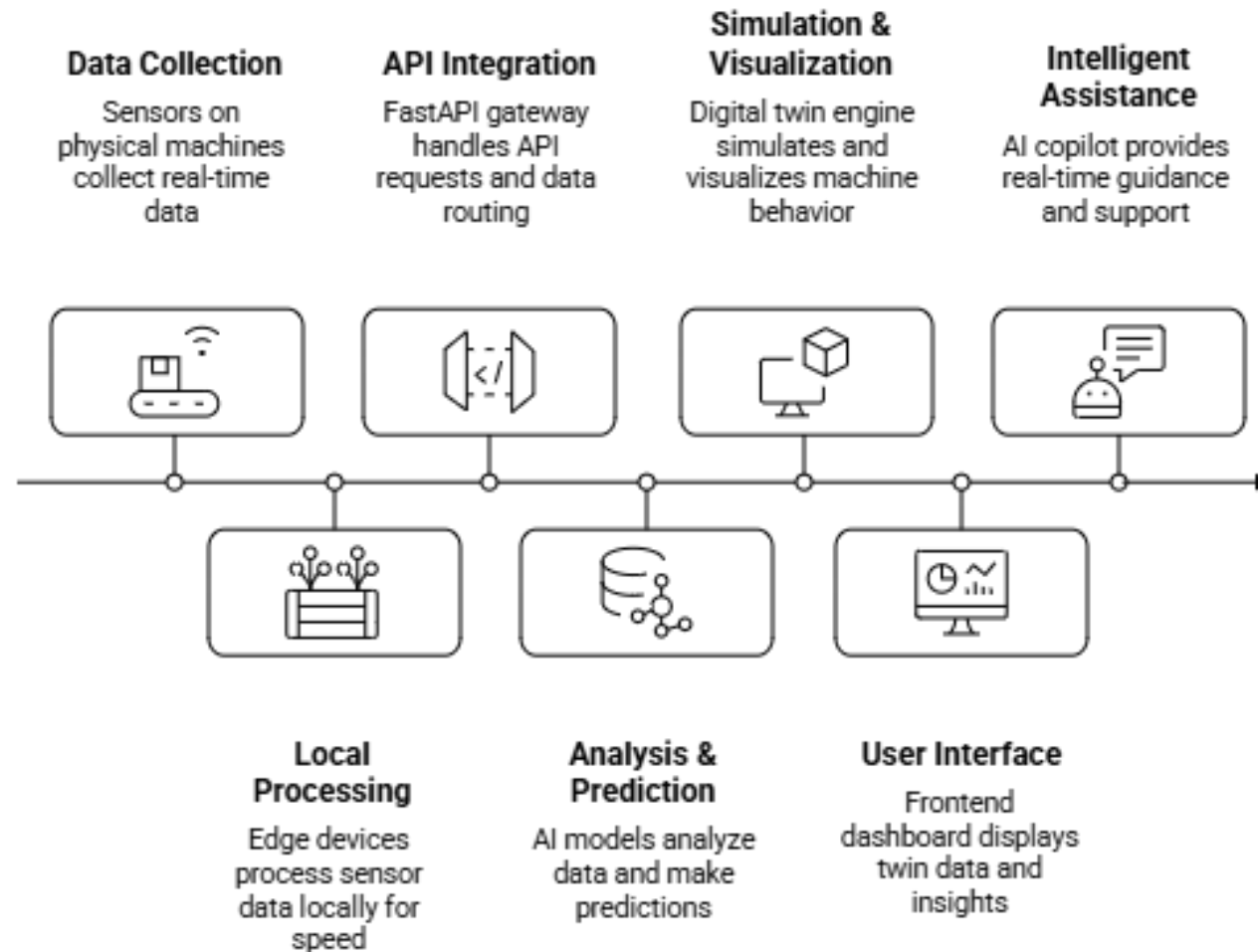


Reports Engine



Smart Alerts

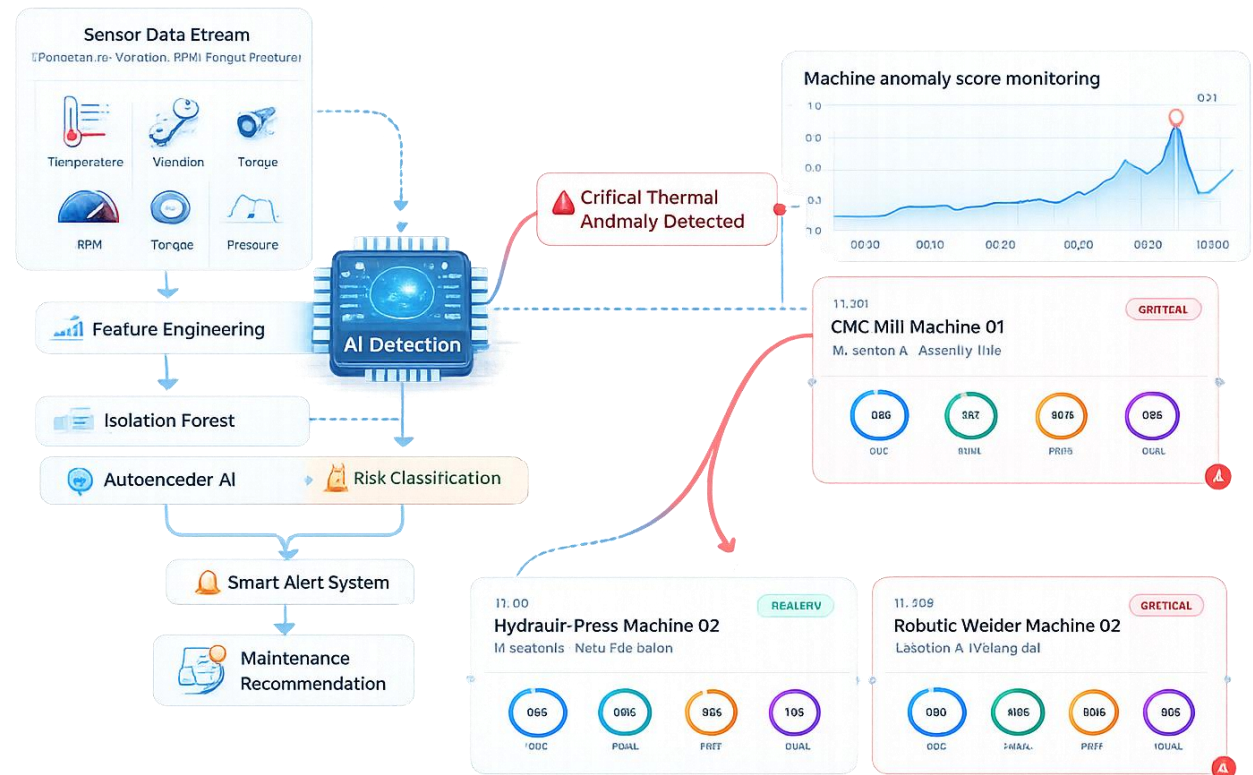
System Architecture



Digital Twin Engine

NexTwin AI creates a real-time virtual replica of the factory floor by continuously synchronizing machine telemetry, operational data, and AI-driven insights.

- Live Machine Monitoring
- Factory Relationship Mapping
- Asset Health Visualization
- Energy Flow Tracking
- OEE Performance Analysis
- Real-Time Decision Support



Predictive Maintenance

Input:

Temperature

Torque

RPM

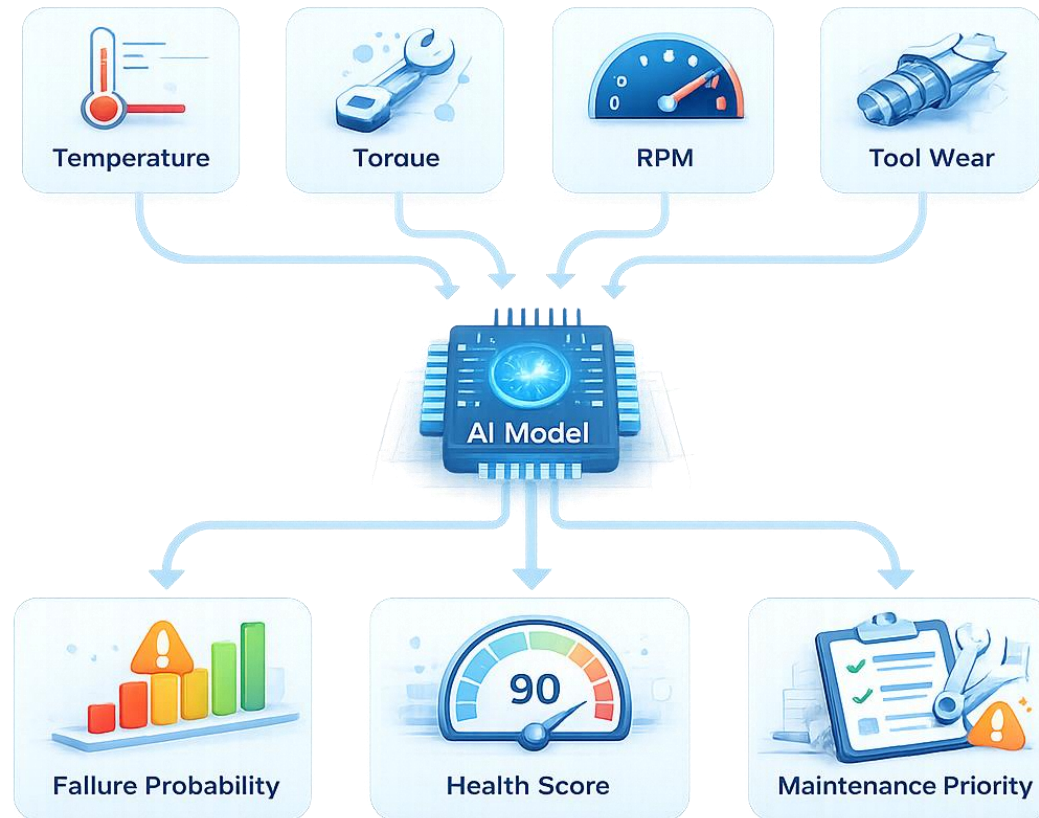
Tool Wear

Output:

Failure Probability

Health Score

Maintenance Priority



AI-Powered Anomaly Detection Engine

Input:

Temperature

Torque

RPM

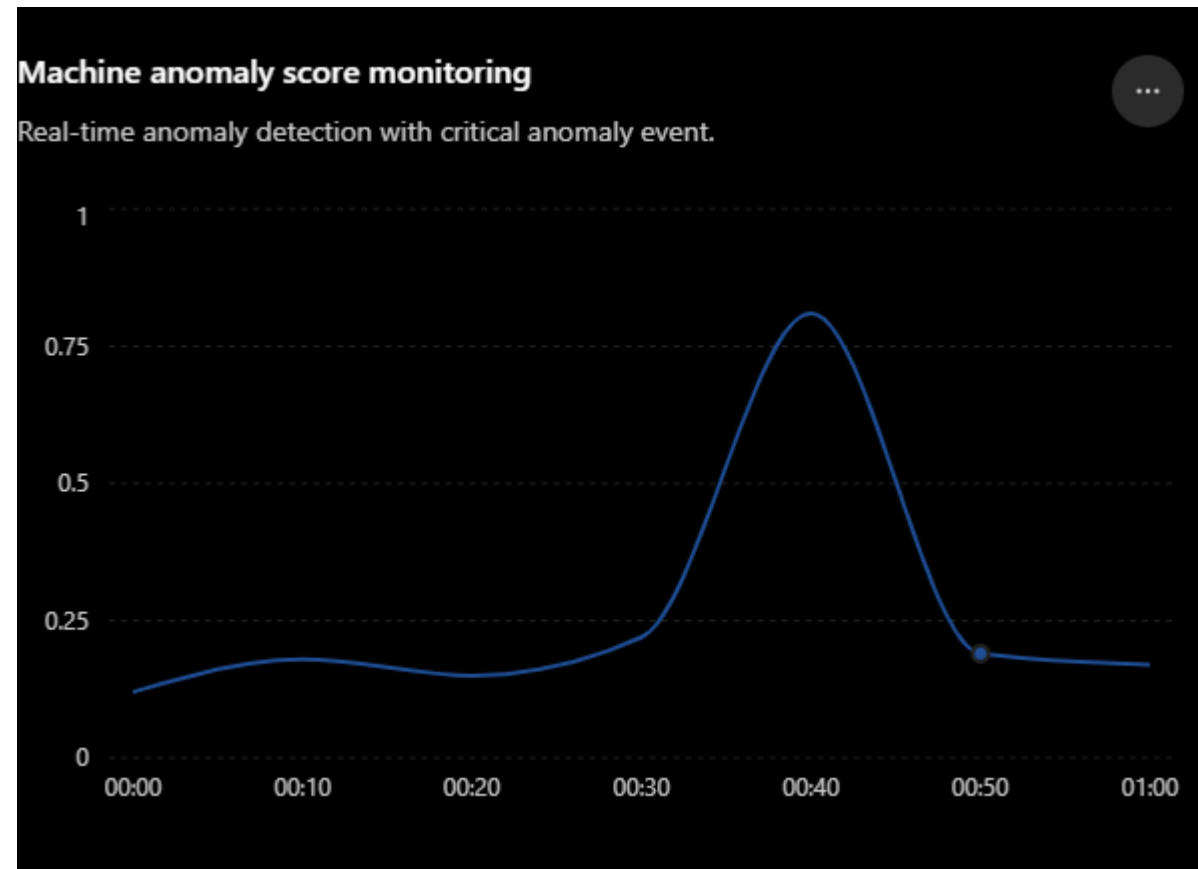
Tool Wear

Output:

Failure Probability

Health Score

Maintenance Priority

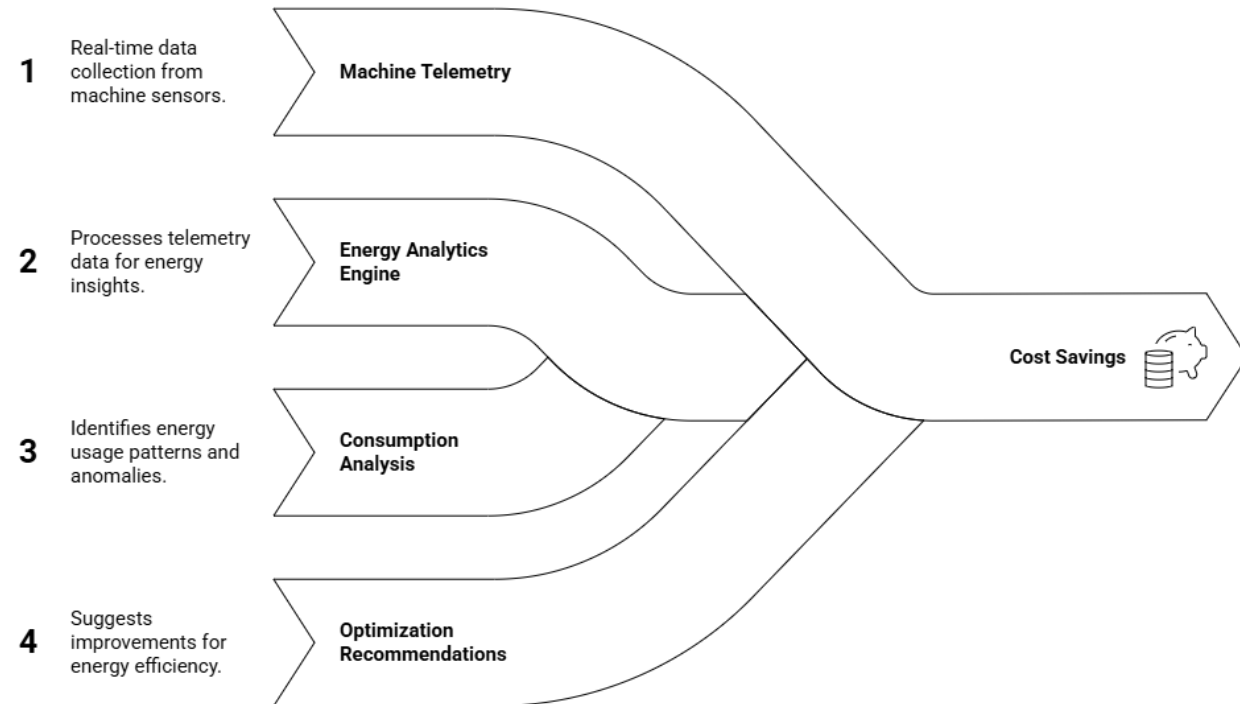


Energy Intelligence Engine

Challenges

- Energy wastage
- Peak load inefficiencies
- High operational cost
- Lack of visibility

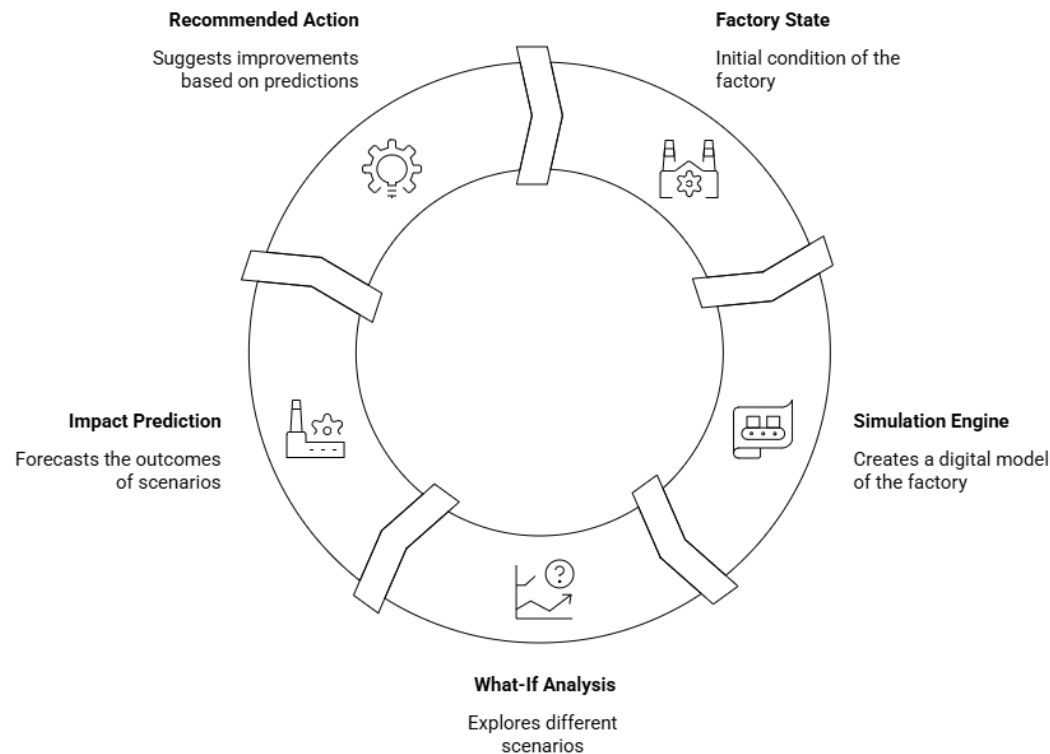
Pathways to Energy Efficiency



Factory Simulation Lab

Test operational changes before implementing them in the real factory.

Factory State Analysis Cycle



Use Cases

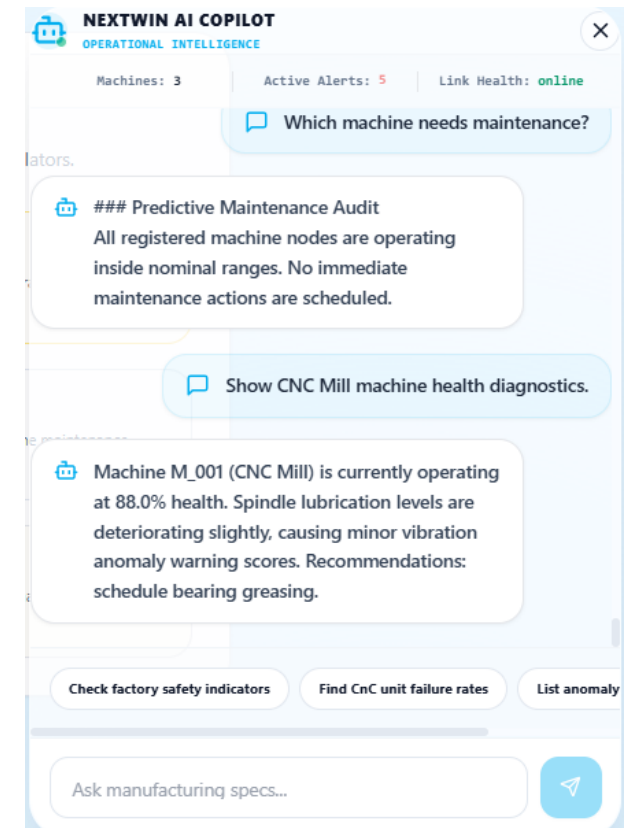
- Machine Failure Simulation
- In
- Bottleneck Prediction
- Throughput Optimization
- Maintenance Scheduling

NexTwin AI Copilot

An intelligent factory assistant that converts industrial data into actionable insights.

AI Copilot

- CNC Mill Machine 01 has a failure risk score of 82%.
- Predicted bearing degradation detected.
- Recommended maintenance window: Within 48 hours.
- Estimated downtime avoidance: 6.5 hours.



Technical Innovation

A unified industrial intelligence platform.

Innovation	Capability
Edge AI	Low-latency processing
Digital Twin	Real-time virtual factory
Predictive Analytics	Failure forecasting
AI Copilot	Natural language insights
Simulation Engine	What-if analysis
Energy Intelligence	Sustainability optimization

Conclusion

Transforming Manufacturing Through AI, Digital Twins & Predictive Intelligence

NexTwin AI is an Edge AI-powered Industrial Digital Twin platform that combines real-time monitoring, predictive maintenance, anomaly detection, energy optimization, factory simulation, and an AI Copilot into a single intelligent ecosystem.

By converting raw industrial data into actionable insights, NexTwin AI enables industries to move from reactive operations to proactive and predictive decision-making.

Thank You

Github : <https://github.com/animesh6532/NexTwinAI.git>